

$$I = \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} = \frac{5}{8}$$

# SEMINAR 2

02.03.2021

Calculati:

1)  $I_m = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^m x \, dx = ?$

2)  $I = \int_0^{\infty} e^{-\frac{1}{2}x^2 - x - 1} \, dx$

3)  $I = \int_0^{\infty} \frac{x^2}{(1+x^4)^2} \, dx$

4) matheia:  $I = \int_0^{\infty} \sin(x^2) \, dx$

SOL:

1)  $I_m = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^m x \, dx =$   
 $= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^{m-1} x \cdot \cos x \cdot \sin x \, dx$

$\sin x \cos x \sin^2 x = t \Rightarrow 2 \sin x \cos x \, dx = dt$

$\Rightarrow I_m = \frac{1}{2} \int_0^1 t^{\frac{m-1}{2}} \cdot (1-t)^{-\frac{1}{2}} \, dt =$

$\Rightarrow I_m = \frac{1}{2} \cdot B\left(\frac{m+1}{2}, \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{\Gamma(\frac{m+1}{2}) \cdot \Gamma(\frac{1}{2})}{\Gamma(\frac{m+1}{2})}$

$\cdot m - \text{par} \Rightarrow I_m = \frac{1}{2} \cdot \frac{\Gamma(\frac{2K+1}{2}) \cdot \Gamma(\frac{1}{2})}{\Gamma(\frac{2K+2}{2})} =$

$= \frac{1}{2} \cdot \frac{\frac{2K-1}{2} \cdot \frac{2K-3}{2} \dots \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma(\frac{1}{2})}{\Gamma(\frac{2K+2}{2})} =$

$= \frac{1}{2} \cdot \frac{1 \cdot 3 \cdot 5 \dots K!}{K!} \cdot \pi =$

$= \frac{(2K-1)!! \cdot \pi}{2^{K+1} \cdot K!} = \frac{1}{2} \cdot \frac{1 \cdot 3 \cdot 5 \dots (2K-1) \cdot \pi}{2 \cdot 4 \cdot 6 \dots (2K)}$

$= \frac{1}{2} \cdot \frac{(2K-1)!!}{(2K)!!} \pi$

$m = \text{impar} \Rightarrow \text{analog}$



$$\cdot \underline{I} = \int_{-1}^{\infty} e^{-\frac{1}{2}x^2} dx$$

$$\int_{-\infty}^{\infty} e^{ax^2+bx+c} dx = \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x+\frac{b}{a})^2 + \frac{b^2}{2a} + c} dx$$

$$= \int_{-\infty}^{\infty} e^{\frac{b^2}{2a} + c} e^{-\frac{1}{2}(x+\frac{b}{a})^2} dx = \dots = \text{Poisson}$$

$$\underline{I} = \int_{-1}^{\infty} e^{-\frac{1}{2}x^2 - x - 1} dx$$

$$-\frac{1}{2}x^2 - x - 1 = -\frac{1}{2}(x^2 + 2x + 2) =$$

$$= -\frac{1}{2}((x^2 + 2x + 1) + 1) =$$

$$= -\frac{1}{2}[(x+1)^2 + 1] =$$

$$= -\frac{1}{2}(x+1)^2 - \frac{1}{2} =$$

$$= -\left(\frac{1}{2}(x+1)^2\right) - \frac{1}{2}$$

$$\underline{I} = \int_{-1}^{\infty} e^{-\left(\frac{1}{2}(x+1)^2\right) - \frac{1}{2}} dx = e^{-\frac{1}{2}} \int_{-1}^{\infty} e^{-\frac{1}{2}(x+1)^2} dx$$

$$u = \frac{x+1}{\sqrt{2}} \Rightarrow x+1 = u\sqrt{2} \Rightarrow x = u\sqrt{2} - 1 \Rightarrow dx = \sqrt{2} du$$

$$x = -1 \Rightarrow u = 0$$

$$x = \infty \Rightarrow u = \infty$$

$$\underline{I} = e^{-\frac{1}{2}} \cdot \sqrt{2} \cdot \int_0^{\infty} e^{-u^2} du = e^{-\frac{1}{2}} \cdot \frac{\sqrt{2} \cdot \sqrt{\pi}}{2} =$$

$$= \frac{\sqrt{2\pi}}{2e^{\frac{1}{2}}} = \frac{\sqrt{2\pi}}{2e^{\frac{1}{2}}}$$

$$\cdot \underline{I} = \int_{-1}^{\infty} e^{-x^2 + 2x - 1} dx = \int_{-1}^{\infty} e^{-(x-1)^2} dx$$

$$x-1 = u \Rightarrow x = u+1 \Rightarrow dx = du$$

$$x = -1 \Rightarrow u = -2$$

$$x = \infty \Rightarrow u = \infty$$



$$I = e^{-x} \cdot \int_0^{\infty} e^{-x} dx = e^{-x} \cdot \frac{1}{1} = \frac{1}{2e^x}$$

Power

$$p(a, b) = \int_0^{\infty} \frac{x^{a-1}}{(1+x)^{a+b}} dx$$

$$I = \int_0^{\infty} \frac{x^2}{(1+x^4)^2} dx$$

$$x^4 = t \Rightarrow x = \sqrt[4]{t} \Rightarrow dx = \frac{1}{4} t^{-3/4} dt$$

$$x=0 \Rightarrow t=0$$

$$x=\infty \Rightarrow t=\infty$$

$$I = \int_0^{\infty} \frac{t^{-1/2}}{(1+t)^2} \cdot \frac{1}{4} t^{-3/4} dt$$

$$= \frac{1}{4} \int_0^{\infty} \frac{t^{-5/4}}{(1+t)^2} dt$$

$$0 = -1 = -\frac{1}{4} = 0$$

$$0 = 2 = 2 = 0$$

$$= \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)} = \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)}$$

$$= \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)} = \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)}$$

$$= \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)} = \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)}$$

$$= \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)} = \frac{1}{4} \cdot \frac{\pi(1/4) \cdot \pi(3/4)}{\pi(1/4) \cdot \pi(3/4)}$$

$$= \pi(1/4) \cdot \pi(3/4) = \pi\sqrt{2}$$

$$= \pi\sqrt{2}$$



• motura  $\int_0^{\infty} \sin(x^2) dx$ .

$$x^2 = t \Rightarrow x = \sqrt{t} \Rightarrow dx = \frac{1}{2\sqrt{t}} dt$$

$$I = \int_0^{\infty} \sin t \cdot \frac{1}{2\sqrt{t}} dt = \frac{1}{2} \int_0^{\infty} \frac{\sin t}{\sqrt{t}} dt$$

(reduced to  $\int_0^{\infty} \frac{\sin x}{x} dx$ )

•  $\int_{-3}^1 \frac{1}{\sqrt{(x+3)^3(1-x)}} dx$

•  $\int_1^3 \frac{1}{\sqrt{(3-x)(x-1)}} dx$

$$t = x-1 \Rightarrow x = t+1 \Rightarrow dx = dt$$

$$I = \int_0^1 \frac{1}{\sqrt{(2-t) \cdot t}} \cdot 2 dt = \int_0^1 \frac{1}{\sqrt{(1-t) \cdot t}} dt =$$

$$= \pi \left( \frac{1}{2}, \frac{1}{2} \right)$$

•  $\int_0^{\infty} (x^2 + 2x + 3) \cdot e^{-x+5} dx =$

$$= e^5 \int_0^{\infty} (x^2 + 2x + 3) \cdot e^{-x} dx =$$

$$= e^5 \left( \int_0^{\infty} x^2 \cdot e^{-x} dx + 2 \int_0^{\infty} x \cdot e^{-x} dx + \int_0^{\infty} e^{-x} dx \right) =$$

$$= e^5 (\pi(3) + 2 \cdot \pi(2) + 3 \cdot \pi(1)) =$$

$$= e^5 (2! + 2 \cdot 1! + 3 \cdot 0!) =$$

$$= e^5 \cdot 7$$



# Serii de Functii

$\sum_{n=1}^{\infty} f_n(x)$   
 ↳ mult de conv.

I. Serii de puteri  $\leadsto \sum_{n=1}^{\infty} a_n x^n$   
 ↳ Raza de convergență?  
 ↳ mult de conv?  
 ↳ Suma seriei?  $\leadsto$  forma specială

$$R = \sqrt[n]{\lim_{n \rightarrow \infty} |a_n|} \quad \Rightarrow \quad R = \frac{1}{\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|} \quad \left( \frac{1}{\infty} = 0, \frac{1}{0} = \infty \right)$$

$\Rightarrow$  seria conv. conv., pt  $x \in (-R, R)$   
 seria div., pt  $x \in (-\infty, -R) \cup (R, \infty)$

Pt  $x = R, x = -R \Rightarrow$  Încercăm un criteriu  
 și obținem 2 serii numerice:

$$\sum_{n=1}^{\infty} (-1)^n \cdot \underbrace{b_n}_{\geq 0} \quad (1)$$

$$\sum_{n=1}^{\infty} b_n \quad (2)$$

$\lim_{n \rightarrow \infty} b_n \neq 0 \Rightarrow (1), (2)$  div.

$= 0 \Rightarrow$  (1) Aplic Leibniz  
 ( $b_n \searrow 0$ )

(2) Aplic crit. de la  
 serie  
 (cresc./decreșc.)

$$\Rightarrow A_x = (-R, R) / [-R, R] / (-R, R] / [-R, R]$$

Ex

$R = ? \quad A_x = ?$

$$\sum_{n=1}^{\infty} (-1)^n \underbrace{\left( 1 + \frac{1}{2} + \dots + \frac{1}{n} \right)}_{a_n} \cdot x^n$$

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} \left( 1 + \dots + \frac{1}{n+1} \right)}{(-1)^n \left( 1 + \dots + \frac{1}{n} \right)} \right|$$



$$= \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \dots + \frac{1}{n+1}}{1 + \frac{1}{2} + \dots + \frac{1}{n}} \Rightarrow \text{S-Criterio:}$$

$$\bullet y_{n+1} - y_n = \frac{1}{n+1} \rightarrow y_n = (y_n)_n \nearrow$$

$$\bullet \lim_{n \rightarrow \infty} y_n = \infty \Rightarrow \text{monoton}$$

$$\bullet \lim_{n \rightarrow \infty} \frac{y_{n+1} - y_n}{y_{n+1}} = 1$$

$$\Rightarrow \rho = 1 \Rightarrow R = 1 \Rightarrow$$

$$\Rightarrow \left\{ \begin{array}{l} \text{abs. conv.}, x \in (-1, 1) \\ \text{div.}, x \in (-\infty, -1) \cup (1, \infty) \end{array} \right.$$

$$x = 1: \sum_{n=1}^{\infty} (-1)^n \left( 1 + \frac{1}{2} + \dots + \frac{1}{n} \right) \cdot 1 \quad (\text{ii})$$

$$x = -1: \sum_{n=1}^{\infty} \left( 1 + \frac{1}{2} + \dots + \frac{1}{n} \right) \quad (\text{iii})$$

$$\bullet \lim_{n \rightarrow \infty} b_n = \infty \Rightarrow b_n$$

$$\Rightarrow b_n \not\rightarrow 0 \Rightarrow$$

$$\Rightarrow (\text{ii}), (\text{iii}) \text{ div.} \Rightarrow$$

$$\Rightarrow \boxed{A_x = (-1, 1)}$$

2.

Det. mult. de converg. serie

$$\sum_{n=1}^{\infty} \left( \frac{6n+1}{6n-5} \right)^n \cdot \left( \frac{x-2}{x} \right)^n, x \in \mathbb{R}^*$$

$$= \sum_{n=1}^{\infty} a_n y^n$$

$$\rho = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \frac{6n+1}{6n-5} =$$

$$= \lim_{n \rightarrow \infty} \frac{6n+1}{6n-5} = 1$$

$$\Rightarrow R = 1 \Rightarrow$$

$$\left\{ \begin{array}{l} \text{abs. conv.}, y \in (-1, 1) \\ \text{div.}, y \in (-\infty, -1) \cup (1, \infty) \end{array} \right.$$

$$\text{div.}, y \in (-\infty, -1) \cup (1, \infty)$$



$$\begin{aligned}
 & \boxed{y=3} \Rightarrow M \left( \frac{23}{63-5} \right) \quad (i) \\
 & \boxed{y=-3} \Rightarrow M \left( \frac{23+1}{63-5} \right) \quad (ii) \\
 & \quad \quad \quad M \left( \frac{63+3}{63-5} \right) = M \left( \frac{66}{58} \right) = \frac{33}{29}
 \end{aligned}$$

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \frac{3}{n} = \lim_{n \rightarrow \infty} \left( \frac{6n+3}{6n-5} \right)^{\frac{1}{n}} \\
 & = \lim_{n \rightarrow \infty} \left( 1 + \frac{8}{6n-5} \right)^{\frac{1}{n}} = e^{\lim_{n \rightarrow \infty} \frac{8}{n(6n-5)}} = e^0 = 1 \neq 0 \Rightarrow \frac{3}{n} \rightarrow 0 \\
 & \Rightarrow (i), (ii) \text{ div} \Rightarrow \\
 & \Rightarrow A_y = (-3, 3) \Rightarrow |y| < 3.
 \end{aligned}$$

$$A_x = \{ x \in \mathbb{R}^* \mid |x-3| < 3 \}$$

..... (prelucrată)

3. clust, de como pt.  $M \left( \frac{33}{3+1} + \frac{33}{3+1} \right) = \frac{33}{2} + \frac{33}{2} = 33$

$$\begin{aligned}
 r &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{33 + (-1)^{n+1}}{33 + (-1)^n} \right| = \dots = 1 \\
 &= \lim_{n \rightarrow \infty} \left| \frac{33 + (-1)^{n+1}}{33 + (-1)^n} \right| = \dots = 1
 \end{aligned}$$

$\Rightarrow R = \frac{1}{1} \Rightarrow$  abs. como.  $y \in (-\frac{1}{4}, \frac{1}{4})$   
 div.  $y \in (-\frac{1}{8}, \frac{1}{8})$   
 $y \in (-\frac{1}{4}, \frac{1}{4})$

$$\begin{aligned}
 & y = \frac{1}{4} \Rightarrow M \left( \frac{33}{3+1} + \frac{33}{3+1} \right) = \frac{33}{2} + \frac{33}{2} = 33 \\
 & = M \left( \frac{1}{50} \right) = \frac{1}{50}
 \end{aligned}$$



$$C_1 = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} M \quad \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} M$$

$$= \frac{1}{\sqrt{2}} M \quad \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} M$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} = 0$$

$$(i): M \left( \frac{1}{\sqrt{2}} \right) = M \frac{1}{\sqrt{2}}$$

$$C_1 = \frac{1}{\sqrt{2}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$= \lim_{n \rightarrow \infty} \left( \frac{1}{\sqrt{2}} \right) + 1 = 1 \in (0, \infty)$$

$\Rightarrow M \frac{1}{\sqrt{2}} = M \frac{1}{\sqrt{2}}$  - via harmonic

$\Rightarrow (i)$  dir.

$$(ii): M \left( \frac{1}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} M$$

$$1 - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \quad M = M + \frac{1}{\sqrt{2}}$$

$$1 - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \quad M = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$$

$$+ M \left( \frac{1}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} M$$

$$M \left( \frac{1}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} M$$

comp. app

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{2}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Get. App.

$$\Rightarrow A \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} A$$

$$\Rightarrow A \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} A$$



SOL:

$$l = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{(-1)^n + 4}} = \lim_{n \rightarrow \infty} \frac{1}{b_n}$$

$$\neq \lim_{n \rightarrow \infty} b_n$$

$$\left. \begin{aligned} b_{2n} &= \frac{1}{5} \\ b_{2n+1} &= \frac{1}{3} \end{aligned} \right\} \Rightarrow l = \max \left\{ \frac{1}{3}, \frac{1}{5} \right\} = \frac{1}{3}$$

$$\Rightarrow R=3 \Rightarrow \left. \begin{aligned} &\text{abs. conv.}, x \in (-3, 3) \\ &\text{div.}, x \in (-\infty, -3) \cup (3, \infty) \end{aligned} \right\}$$

$$x=3 \Rightarrow \sum_{n=1}^{\infty} \left( \frac{3}{(-1)^n + 4} \right)^n \quad (i)$$

$$x=-3 \Rightarrow \sum_{n=1}^{\infty} (-1)^n \cdot \left( \frac{3}{(-1)^n + 4} \right)^n \quad (ii)$$

$$\cdot \lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \left( \frac{3}{(-1)^n + 4} \right)^n \neq$$

$$\left. \begin{aligned} \lim_{n \rightarrow \infty} c_{2n} &= 0 \\ \lim_{n \rightarrow \infty} c_{2n+1} &= 1 \end{aligned} \right\} \Rightarrow c_n \not\rightarrow 0 \Rightarrow$$

$$\Rightarrow (i), (ii) \text{ div.} \Rightarrow$$

$$\Rightarrow A_x = (-3, 3)$$

5. Det. mult. de conv. a seriei de fct.  $\sum_{n=0}^{\infty} 2^n \cdot \sin \frac{x}{3^n}$ ,  $\forall x \in [-2, 2]$ . Stabilitate de conv. e uniformă.

SOL:

$$\text{Fie } f_n: [-2, 2] \rightarrow \mathbb{R}, f_n(x) = 2^n \cdot \sin \frac{x}{3^n}$$

$$|f_n(x)| = \left| 2^n \cdot \sin \frac{x}{3^n} \right| \quad (*)$$

$$\text{Avem } |\sin x| \leq |x|, \forall x \in \mathbb{R}$$

$$\Rightarrow |f_n(x)| \leq \left| 2^n \cdot \frac{x}{3^n} \right| = \left( \frac{2}{3} \right)^n \cdot |x| = \left( \frac{2}{3} \right)^n \cdot 2$$



Da seia numerau

seie geometria

(3) Seia dată e unist. comut. si  
 $A = [-2, 2]$ .

$\sum_{n=1}^{\infty} x^n =$  comut, cu suma  $\frac{x}{1-x}$ ,  $x \in (-1, 1)$   
 $\sum_{n=0}^{\infty} x^n =$  comut, cu suma  $\frac{1}{1-x}$ ,  $x \in (-1, 1)$

Ex:

$R = ?$   
 $A_x = ?$   
 $S(x) = ?$

pt. serie  $\left\{ \begin{array}{l} \textcircled{1} \sum_{n=1}^{\infty} n \cdot x^n \\ \textcircled{2} \sum_{n=1}^{\infty} n(n+1) \cdot x^n \\ \textcircled{3} \sum_{n=1}^{\infty} n^2 \cdot x^n \end{array} \right.$

Sol:

Evident  $R=1$ ,  $A_x = (-1, 1)$

Sol:

(1) Stim  $\sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$ ,  $x \in (-1, 1)$   
 $\Rightarrow \sum_{n=1}^{\infty} (x^n)' = \left( \frac{x}{1-x} \right)'$ ,  $x \in (-1, 1)$   
 $\Rightarrow \sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$ ,  $x \in (-1, 1)$

$\Rightarrow \sum_{n=1}^{\infty} n \cdot x^n = \frac{x}{(1-x)^2}$ ,  $x \in (-1, 1)$

(2) Stim  $\sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$ ,  $x \in (-1, 1)$   
 $\Rightarrow \sum_{n=1}^{\infty} x^{n+1} = \frac{x^2}{1-x}$   
 $\Rightarrow \sum_{n=1}^{\infty} (n+1) \cdot x^n = \left( \frac{x^2}{1-x} \right)' = \frac{2x(1-x) + x^2}{(1-x)^2}$   
 $= \frac{2x - x^2}{(1-x)^2}$



$$\Rightarrow \sum_{n \geq 1} (n+1)n x^{n-1} = \left( \frac{2x - x^2}{(1-x)^2} \right)' =$$

$$= \frac{(2-2x)(1-x)^2 + (2x-x^2) \cdot 2(1-x)}{(1-x)^4} =$$

$$= \frac{(2-2x)(1-x) + (2x-x^2) \cdot 2}{(1-x)^3} =$$

$$= \frac{2-4x+2x^2+4x-2x^2}{(1-x)^3} = \frac{2}{(1-x)^3}$$

$$\Rightarrow \sum_{n \geq 1} (n+1) \cdot n \cdot x^n = \frac{2x}{(1-x)^3}, \forall x \in (-1, 1) \setminus \{0\}$$

$$(b) : \sum_{n \geq 1} n^2 \cdot x^n = \sum_{n \geq 1} (n^2 + n - n) \cdot x^n =$$

$$= \sum_{n \geq 1} n(n+1) \cdot x^n - \sum_{n \geq 1} n \cdot x^n =$$

$$= \frac{2x}{(1-x)^3} - \frac{x}{(1-x)^2}, \forall x \in (-1, 1) \setminus \{0\}$$

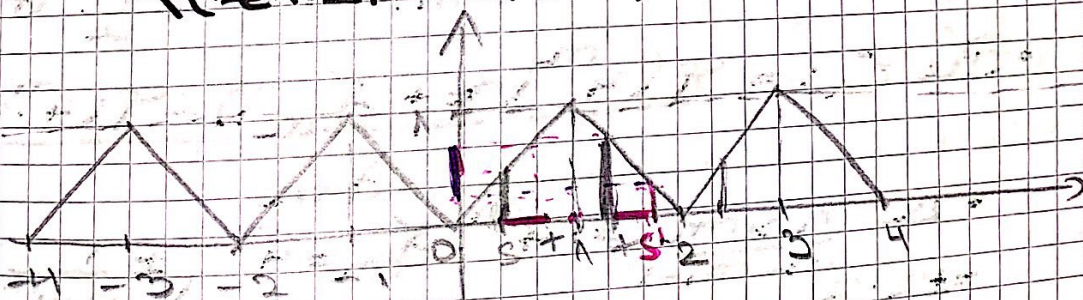
Ex: Dem. că  $\exists f: \mathbb{R} \rightarrow \mathbb{R}$  continuă, dar care nu e derivabilă în niciun punct.

SOL:

Fie  $f: [-1, 1] \rightarrow \mathbb{R}, f(x) = |x|$ .

Existăm  $\varphi$  pe  $\mathbb{R}$  astfel:

$$\varphi(x+2) = \varphi(x), \forall x \in \mathbb{R}$$



Obs că  $\forall s, x \in \mathbb{R}$  avem că  $|f(s) - f(x)|$

$$\leq |s - x| \quad (*)$$

de  $|s - x| \geq 1 \Rightarrow (*)$  evident.



Obs că  $f$  cont. pe  $\mathbb{R}$  și cum  $f(x) \in [0, 1]$   
 $\forall x \in \mathbb{R}$ , considerând seria de funcții:

$$\sum_{n \geq 0} \left(\frac{3}{4}\right)^n \cdot f(4^n \cdot x),$$

vom adăuna că ea este maj. de v. n. a  
 numerică  $\sum_{n \geq 0} \left(\frac{3}{4}\right)^n$  de term. poz., geom.

(X) seria  $\sum_{n \geq 0} \left(\frac{3}{4}\right)^n \cdot f(4^n \cdot x)$  conv. <sup>geom.</sup>  
 unif. pe  $\mathbb{R}$ .

Fie  $g: \mathbb{R} \rightarrow \mathbb{R}$  suma sa,  $g(x) = \sum_{n \geq 0} \left(\frac{3}{4}\right)^n \cdot f(4^n \cdot x)$ , deci  $g$  cont. pe  $\mathbb{R}$ .

Dem. că  $g$  nu este deriv. în  
 niciun punct din  $\mathbb{R}$ :

$$\Leftrightarrow \forall x \in \mathbb{R}, \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \neq \text{un nr.}$$

• Fie  $x \in \mathbb{R}$  fixat și  $m \in \mathbb{N}$ .

Fie  $\delta_m := \frac{1}{2} \cdot 4^{-m} = \pm \frac{1}{2 \cdot 4^m}$ , unde  
 termenul este ales at. că NU există  
 niciun nr. ambig ambu nr.

$$4^m x \text{ și } 4^m (x + \delta_m) = 4^m x \pm \frac{1}{2} \quad (**)$$

• Fie  $\gamma_m := \frac{f(4^m (x + \delta_m)) - f(4^m x)}{\delta_m}$ .

$$\begin{aligned} \text{• De } m > m_1 &\Rightarrow 4^m \cdot \delta_m = 4^m \left( \pm \frac{1}{2 \cdot 4^m} \right) = \\ &= \pm 2 \cdot \frac{1}{4^{m-m_1-1}} \text{ ambig, } \underline{\text{par}} \Rightarrow \end{aligned}$$

$$\Rightarrow \gamma_m = 0 \quad (1)$$

• De  $0 \leq m < m_1$  (\*)

$$\Rightarrow |\gamma_m| = \left| \frac{4^m x + 4^m \delta_m - 4^m x}{\delta_m} \right| = 4^m \quad (2)$$



• De  $m = m$  :  
 $|x_m| = \frac{\varphi(4(x + \delta_m)) - \varphi(4x)}{\delta_m} \begin{matrix} \text{nu exista niciun} \\ \text{indrag} \end{matrix}$   
 $= \frac{1}{2\delta_m} = 4^m \quad \text{desen.}$

[illegible]